

EE115B Project Environment Setup Tutorial

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Part 1 Introduction

In EE115B, we designed a simple project to allow everyone to experience the process of designing digital circuits using Verilog (a kind of HDL, Hardware Design Language). This document helps everyone set up an extremely lightweight Verilog integrated development environment on their own computers.

Icarus Verilog is a lightweight, free and open-source Verilog compiler implemented in C++. It was developed by Stephen Williams and is licensed under the GNU GPL license. The installation package includes GTKWave support for compiling and simulating Verilog/VHDL files. It operates in a command-line manner, similar to the `gcc` compiler. Through testbench files, corresponding simulation waveform data files can be generated, and the simulation waveform diagrams can be viewed through the built-in GTKWave.

Part 2 Install Icarus Verilog

Some References that you might find useful:

- Video tutorial for installation: [Link](#)
- Installation Tutorial [Link](#)
- Video tutorial for `hello_world` project: [Link](#)

2.1 Installation for Windows

1. Install the software at <https://bleyer.org/icarus/>, select `iverilog-v11-20201123-x64_setup.exe`
2. Click on `iverilog-v11-20201123-x64_setup.exe` and select **Full installation**. Other versions should work as well.
3. Select “Add executable folder(s) to the PATH”
4. Ready to install
5. Open `cmd` terminal and check the installation address to make sure the installation is successful:

```
where iverilog
where vvp
where gtkwave
```

2.2 Installation for Linux

Open terminal and enter:

```
sudo apt-get install iverilog
sudo apt-get install gtkwave
```

Check whether the installation is successful:

```
which iverilog
which vvp
which gtkwave
```

2.3 Installation for MacOS

Using MacPorts:

```
sudo ports -d -v install iverilog
sudo ports -d -v install gtkwave
```

Or using Homebrew:

```
brew install icarus-verilog
brew install caskroom/cask/gtkwave
```

Part 3 How to Compile Verilog Files and Check Waveform

1. Open the location of the homework folder on the terminal and enter the corresponding folder of the question.
2. Finish your code and testbench in `.v` and `_tb.v` files respectively. For those testbenches which are given, **DO NOT MODIFY** the `xxx_tb.v` file unless it is required by homework requirement.
3. Enter the iverilog command to generate a `.vvp` file:

```
iverilog -o xxx_tb.vvp xxx_tb.v
```

4. Enter the vvp command to generate a `.vcd` file:

```
vvp xxx_tb.vvp
```

5. Enter the gtkwave command to check your design's waveform:

```
gtkwave xxx_tb.vcd
```

6. Select the signal you want to check the corresponding waveform.
7. Try steps 1 and 3–6 in the folder `hello_world`, and get familiar with Verilog compilation yourself!
8. Expected result:

